

Perfect. I've absorbed the dump 👍

Below is your **organized, cleaned, and structured version** — same ideas, zero loss, just made *expo-ready* and *team-understandable*. Think of this as your **single source of truth**.

Project Overview (Smart Education)

Working Idea:

An **AI-powered Smart Education Platform** focused **exclusively on VTU university students**, helping them manage studies, exams, notes, PYQs, and performance using agentic AI — from daily study to final semester exams.

The system operates in **two core modes**:

- **Study Mode** → daily learning, tasks, notes, revisions
 - **Exam Mode** → PYQ analysis, question prediction, grading, performance tracking
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Target Users

- **Primary:** VTU University students (initial phase only)
 - **Secondary (future):** Other Indian technical universities
 - **User Types:**
 - o Free users
 - o Standard users
 - o Premium users
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Core Modes

1. Study Mode

Focused on **continuous learning & productivity**:

- Daily study task planning
- Assignment assistance
- Revision sessions
- Smart notes generation
- Lecture transcription → topic mapping
- Flashcards & mind maps
- Live AI-assisted document editing

2. Exam Mode

Focused on **scoring & exam readiness**:

- PYQ-based preparation
 - VTU-style answers (2M / 8M / 10M)
 - Question importance & prediction
 - Mock tests & grading
 - IA & Semester exam tracking
 - Weakness analysis & improvement suggestions
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User Onboarding & Profile

Each user must sign up with **academic details**, including:

- College name
- University (VTU)

- Course & branch
- Subjects
- Modules / Units
- AECs (Ability Enhancement Courses)

This enables **subject-wise AI personalization**.

Document & Knowledge Management System

Upload & Organization

Users can upload:

- PDFs (notes, PYQs, worksheets)
- Images (question papers)
- Text
- YouTube links
- Lecture recordings

AI will:

- Auto-rename files (clean titles)
- Generate descriptions
- Classify document type (PYQs / Notes / Important Qs / Worksheets)
- Store under:
Subject → Module → Category

Study Kits & Sessions

- Users create **custom study kits**
- Select documents for a **study session**
- AI can **only access selected documents**

- Live chatbot can reference documents using @filename
 - Search across entire database (Standard & Premium only)
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AI Capabilities (Feature-wise)

Learning & Content Creation

- Summarize documents & YouTube videos
- Convert content into:
 - VTU-style Q&A
 - 2M / 8M / 10M answers
- Generate:
 - Mind maps
 - Slides
 - Flashcards (2-mark theory only)
 - Podcasts using TTS

Lecture Intelligence

- Transcribe recorded lectures
- Understand lecturer instructions
- Map lecture content to syllabus & documents
- Auto-create tasks and reminders

Exam Intelligence

- Solve uploaded questions (VTU marking scheme)
- Paper grading AI (evaluate written answers)
- OCR + voice-based repetitive answering practice
- Important question prediction using PYQ analysis

- Question importance & probability rating

Performance Tracking

- IA tracker
 - Semester exam tracker
 - Marks, question papers & feedback
 - Strength/weakness analysis
 - Lecturer instructions storage
 - Personalized improvement roadmap
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Library & Database Strategy

Centralized Knowledge Library

- All documents entering the system are:
 - Stored with metadata
 - Categorized properly
 - Added to a **global catalog**
- User-uploaded documents can be reused **only with user consent**

External Data Sources

- Scraped from VTU-related sites (e.g., Backbenchers, VTU Circle, etc.)
 - Stored in external database
 - AI accesses **only verified documents** (no hallucination)
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Monetization Model (API-based)

Using **Kimi K 2.5 (open-source)** + OpenRouter for model flexibility.

Plans:

Free

- User brings their own API key

Standard

- Platform AI credits (standard rate limits)
- User can also use their own API key
- Access to database search & saved content

Premium

- Platform-provided API keys
 - Higher rate limits
 - Full database access
 - Advanced exam tools
 - Personal knowledge vault
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Technical Architecture

Application

- Web application
- PWA support (planned)

Frontend

- React
- TypeScript

Backend & Storage

- Supabase → user auth + temporary uploads
- External database → long-term documents & notes

- Cloudflare R2 → hosting & large-scale storage
- Focus on **low cost** + **high bandwidth**

AI & Dev Tools

- Kimi K 2.5 (core model)
- OpenRouter (model switching)
- Claude CLI
- ChatGPT Codex
- Google AI Studio
- Antigravity, Emergent

Development Workflow

- Git-based version control
 - Mandatory CHANGELOG.md
 - Clear rules on:
 - What can be modified
 - What must not be touched
 - AI-assisted coding with strict boundaries
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What This Is (In One Line)

A **VTU-focused AI study companion** that manages learning, documents, exams, and performance — from first lecture to final exam.

You're in a **very strong Smart Education lane**.

Next logical steps (when *you* say so):

- Convert this into **Project Expo abstract**
- Reduce to **problem–solution–innovation**

- Map features → **Special Awards**
- Decide **MVP vs future scope**

Just tell me the next move.